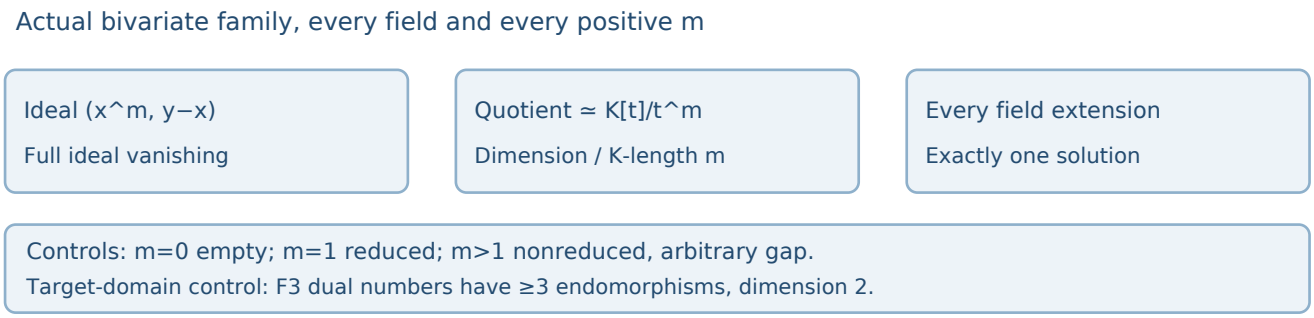
Actual solutions and coordinate-algebra length

Generic formal bridge. For any multivariate ideal I over a field K, Lean constructs an equivalence between its actual solutions over any field extension L and K-algebra maps K[variables]/I → L. When the quotient is finite dimensional, the solution set is finite and its cardinality is at most the quotient dimension, which equals its K-module length. Nilpotents in the source algebra are permitted; no reducedness or basis premise is supplied. Signed standing unverified. [pcc:unverified]



Actual family. For every field K and every natural m, the coordinate algebra of the actual two-variable ideal (x^m, y−x) is explicitly isomorphic to K[t]/(t^m). Both compositions are proved on actual quotient coordinates/root. A monic power basis gives dimension and scalar-module length m. The actual coordinate x has mth power zero; for m>0 its previous power is nonzero. For m>1 the actual quotient is nonreduced. [pcc:unverified]

Support and controls. The full ideal vanishing condition, over every field extension, is equivalent to x^m=0 and y=x. For m>0 this gives exactly the origin and solution cardinality one. Lean proves an arbitrarily large dimension with unchanged support and directly negates universal support=dimension. At m=0 the solution set is empty and dimension/length zero; at m=1 the actual quotient is reduced and isomorphic to K. These are formal controls over every field, without solver enumeration. [pcc:unverified]

Why the target condition matters. The checked homomorphism bound requires a domain target. For the actual finite F3 algebra A=F3[t]/t², scaling its nonzero nilpotent root by each field element gives an injective family of endomorphisms. Lean proves card End(A)≥3 while dim A=2, and directly negates the bound with this nonreduced target. This tests deleting a stated hypothesis; it does not refute the qualified theorem. [pcc:unverified]

Sources checked. Original Diem (2011), Proposition 2.5, DOI10.1112/S0010437X10005075; existing actual bivariate multiplicity/JV basis-length packets; pinned Mathlib quotient lifts/ext, evaluation uniqueness, algebra-map independence/cardinality, monic AdjoinRoot power bases and module length. Exact Stacks09HS/0CKM statements and proofs were read and retained: they concern field extensions and are narrower than the actual checked ring-algebra statement. Neither citation annotations nor an irreducibility docstring replace actual formal hypotheses. No full-paper read or novelty claim. [pcc:unverified]

Validation. Whole gate 44 declarations / 0 failures / 0 trusted native exceptions. Lean4.35.0 / Mathlib56a00585…; complete closed binding, custom source/config and artifact pins retained. All failed/successful source/log/exit records and the full gate are losslessly archived. Only propext/Classical.choice/Quot.sound; no holes, custom axioms, native evaluator, raised limits, new global flags or waivers. Exact PCC replay is separate; signed standing unverified. [pcc:unverified]

Remaining geometry and execution. This proves scalar-module length accounting for finite affine coordinate algebras. Scheme-cycle/local intersection multiplicity identification, isolated points within positive dimensional components, the equal-multidegree projective n!d^n bound and Rojas/C refinement remain open. Neither Diem's qualified theorem nor the corpus node is refuted. Existing 16 msolve calibration calls remain unchanged; no new solver experiment, paid compute or agent upload. KB unavailable; bounded source/formal fallback recorded. All fourteen original requirements retain their precise open scopes. [pcc:unverified]